Lab 1

dvdrental/postgres@PostgreSQL

Query | Query History

```sql
SELECT relname as object_name,
    pg_size_pretty(pg_table_size(fed.oid)) as data_size,
    pg_size_pretty(pg_indexes_size(fed.oid)) as index_size,
    pg_size_pretty(pg_total_relation_size(fed.oid)) as total_size
FROM pg_class fed
JOIN pg_namespace n ON n.oid = fed.relnamespace
WHERE fed.relkind = 'r' AND n.nspname = 'public'
ORDER BY pg_total_relation_size(fed.oid) DESC;
```

Data Output | Messages | Notifications

Showing rows: 1 to 15 Page No: 1 of 1

	object_name name	data_size text	index_size text	total_size text
1	rental	1232 kB	1120 kB	2352 kB
2	payment	896 kB	920 kB	1816 kB
3	film	736 kB	216 kB	952 kB
4	film_actor	272 kB	216 kB	488 kB
5	inventory	232 kB	208 kB	440 kB
6	customer	96 kB	112 kB	208 kB
7	address	88 kB	64 kB	152 kB
8	city	64 kB	48 kB	112 kB
9	film_category	72 kB	40 kB	112 kB
10	actor	40 kB	32 kB	72 kB
11	store	8192 bytes	32 kB	40 kB
12	staff	16 kB	16 kB	32 kB
13	language	8192 bytes	16 kB	24 kB

Total rows: 15 Query complete 00:00:00.284 CRLF Ln 8, Col 47

Lab 2

/* current size */

```sql
/* current size */
SELECT pg_size_pretty(pg_indexes_size('payment'));
/* create heavy index on payment_date */
CREATE INDEX idx_payment_date_long
ON payment(payment_date, amount, customer_id);
/* check size after create index */
SELECT pg_size_pretty(pg_indexes_size('payment'))
```

Data Output | Messages | Notifications

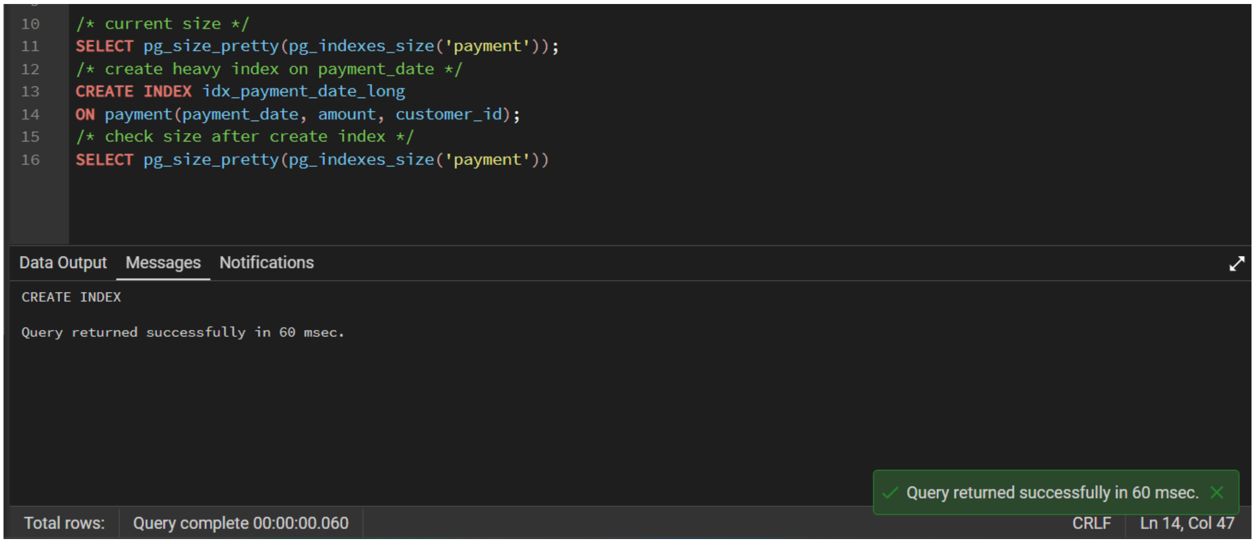
Showing rows: 1 to 1 Page No: 1 of 1

	pg_size_pretty text
1	920 kB

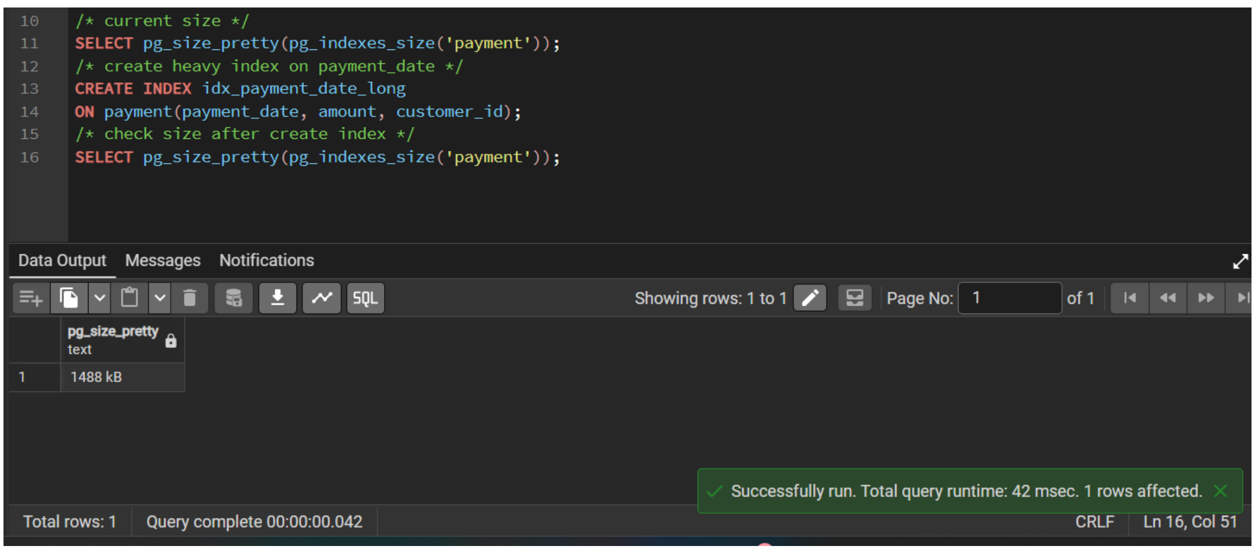
Successfully run. Total query runtime: 70 msec. 1 rows affected.

Total rows: 1 Query complete 00:00:00.070 CRLF Ln 11, Col 51

/* create heavy index on payment_date */



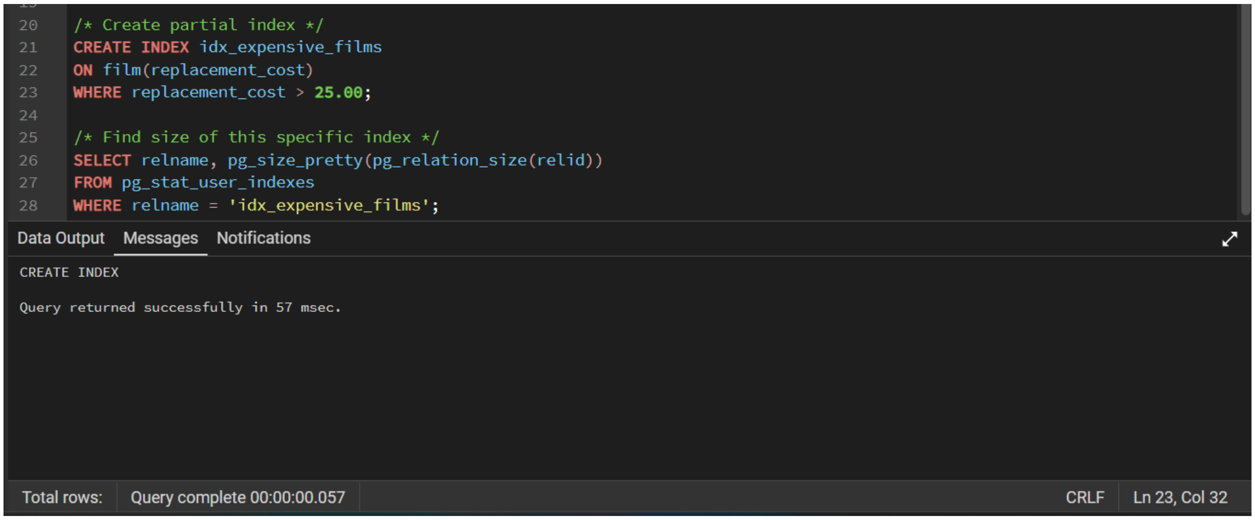
/* check size after create index */



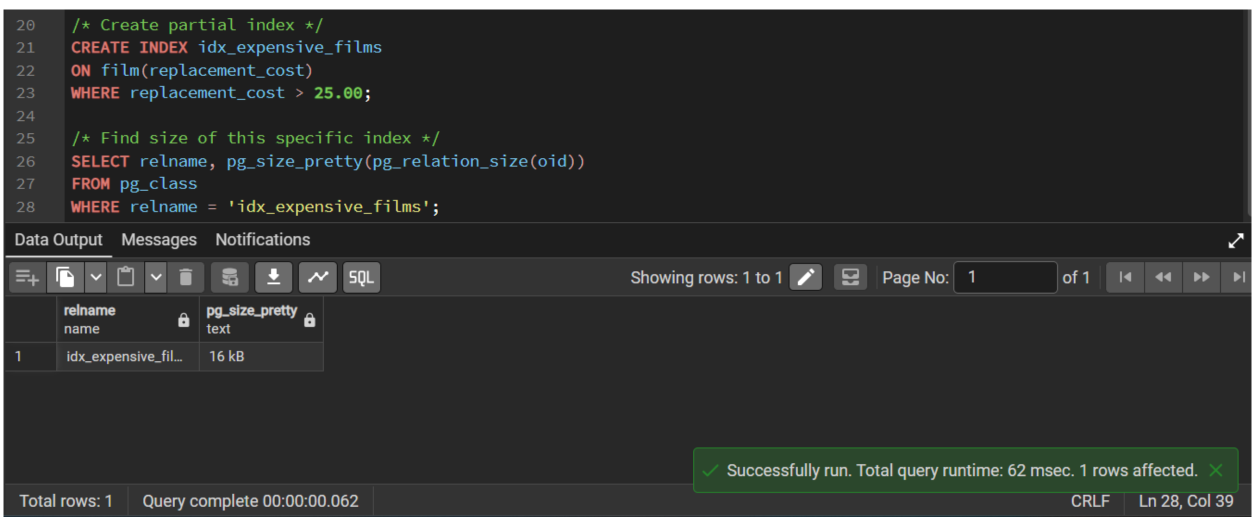
Discussion: New index consumes (1488 – 920)/8 = 71 pages

Lab 3

/* Create partial index */



/* Find size of this specific index */



Task: 16kB of this specific index is very small compare to 568kB of hypothetical full index on replacement_cost.